

DROS-PGM: A Deterministic Post-Compromise Execution Containment Substrate for Autonomous AI Workloads (v2.0)

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Abstract—In modern autonomous AI agent workloads where agents obtain legitimate credentials and invoke consequential tools, traditional application-layer perimeters suffer from fundamental failure modes. Conventional defenses rely on probabilistic semantic guardrails or coarse-grained operating system sandboxes under the assumption of preventing compromise. However, once internal dialogue or interpreter environments succumb to indirect prompt injection, attackers inevitably inherit valid credentials and execute irreversible physical state changes. This paper presents DROS-PGM, a post-compromise execution containment substrate operating at the binary execution control plane. Architecturally, the C-ABI / FFI boundary manages policy evaluation and principal capability attribution, while OS kernel hooks enforce mandatory authorization checks over explicitly instrumented operation classes X_{covered} . PGM decouples application principal identity from binary execution authority, maintaining the formal containment invariant via sub-microsecond ($P50 = 353$ ns) lock-free evaluation and atomic RCU state-pointer swaps. To rigorously evaluate boundary robustness without self-witness circularity, we introduce the PGM-VEP Five-Tier Progressive Falsification Methodology (V1–V5), encompassing attack-equivalent baselines, adaptive white-box state search, negative control meta-verification (5/5 injected flaw detection and 100/100 mutant kill score), a four-stage decoupled ground-truth oracle pipeline ($O_I \rightarrow O_A \rightarrow O_E \rightarrow O_P$), and cross-environment replication across Linux x86_64, ARM64, and Windows. Across 118,355 total executions (68,355 adversarial + 50,000 benign, BFDR = 0/50,000), zero unauthorized executions or state drifts were observed within explicitly instrumented boundaries. We report these guarantees as empirical invariants over the evaluated state space and establish an open counterexample registry to support ongoing adversarial falsification.

Index Terms—Execution Containment Substrate, Post-Compromise Security, Kernel Enforcement Boundary, Dynamic Capability Revocation, Progressive Adversarial Falsification, Decoupled Oracles.

I. PROBLEM DEFINITION: COMPROMISE DOES NOT ENTAIL AUTHORITY

Autonomous AI Agent architectures routinely operate with elevated privileges. In these environments, security architectures encounter a fundamental Semantic-Kernel Paradox:

- 1) **Semantic Application Middleware (High Semantics, Zero Determinism):** Prompt firewalls and JSON

schema validators operate purely at the natural language plane, lacking physical execution containment.

- 2) **Operating System Sandboxes (High Determinism, Zero Semantics):** Traditional kernel mechanisms enforce binary constraints but lack application semantic context, failing to distinguish between legitimate and compromised agents within the same process.

We formalize the core research thesis: **Compromise of an application principal does not entail inheritance of unrestricted execution authority** ($C_A \not\Rightarrow C_E$).

II. DUAL BOUNDARIES & ARCHITECTURAL PLACEMENT

The system establishes a strict separation of concerns via a Dual-Boundary Model:

- **Boundary 1 (C-ABI / FFI Attribution Boundary):** Manages policy evaluation and principal capability attribution, evaluating in 353 ns ($P50$) via $O(1)$ SIMD bitmasks.
- **Boundary 2 (OS Kernel Mandatory Enforcement Boundary):** Intercepts operations over the explicitly instrumented operation classes X_{covered} .

$$X_{\text{covered}} = X_{\text{fs}} \cup X_{\text{proc}} \cup X_{\text{net}} \cup X_{\text{ipc}} \quad (1)$$

Non-Claim Clause: PGM does not claim syscall-complete mediation over uninstrumented OS kernel paths.

III. FORMAL INVARIANTS & SECURITY CLAIM MATRIX

A. Formal Invariants

- 1) **Non-Inheritance of Authority Axiom:** $C_A \not\Rightarrow C_E$
- 2) **Covered Containment Invariant:**

$$\forall x \in X_{\text{covered}}, \quad \text{Auth}_E(x) = \text{DENY} \implies \text{Exec}_{\text{unauth}}(x) = 0 \wedge \Delta S_{\text{Sobs}} = 0 \quad (2)$$

- 3) **Ghost Syscall Measurement Oracle:**

$$G = \#\{x \in X_{\text{covered}} \mid \text{Auth}_E(x) = \text{DENY} \wedge \text{SyscallObserved}(x) = 1\} \quad (3)$$

- 4) **Linearization Semantics of Revocation:** Dynamic revocation takes effect deterministically for all subsequent calls at $T_{\text{swap}} = 420$ ns.

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TABLE I
DROS-PGM SECURITY CLAIM BOUNDARY MATRIX

Claim Category	Statement & Bounded Scope	Epistemic Status
Claimed	In X_{covered} , DENY operations produce zero unauthorized execs	Architectural Invariant
Claimed	Revocation takes effect immediately at T_{swap}	Linearization Semantics
Observed	Observed ghost syscall count $G = 0$	Empirical Fact
Observed	Observed state drift $\Delta S = 0$ over $\mathcal{S}_{\text{obs}}$	Multi-Oracle Fact
Not Claimed	Global syscall-complete mediation	Explicit Non-Claim
Not Claimed	Defense against kernel rootkits / MMU bypass	Host TCB Assumption
Not Claimed	Upstream semantic bitmask correctness	Upstream Duty

IV. THREAT MODEL: POST-COMPROMISE ASSUMPTION ($C_A = 1$)

- **In-Scope Attacker:** Full application compromise ($C_A = 1$), in-memory credential access, multi-agent confused deputy evasion.
- **TCB Assumptions:** PGM binary gate, CPU Ring 0/MMU, OS LSM drivers, immutable audit logs.

V. PGM-VEP VALIDATION METHODOLOGY (V1–V5)

Enforcing the No-Self-Witness Principle ($O_I \neq O_A \neq O_E \neq O_P$), we structure evaluation across five tiers:

TABLE II
PGM-VEP METHODOLOGY TIERS TO ASSET MAPPING

Tier	Executable Harness	Evidence Asset Path
V1	run_benchmark.py	CYBERMES_REPORT.md
V2	stress_test_acc.py	DROS_24H_Report.md
V3	conformance_test.py	conformance_rep.json
V4	replay.py	COMPARATIVE_REP.md
V5	docker-compose.yml	benchmark_sum.json

VI. EXPERIMENTAL RESULTS & METRIC ANALYSIS

A. Evaluation Accounting Matrix

TABLE III
EVALUATION ACCOUNTING MATRIX

Tier	Configuration & Architecture	Execs	CE	Status
V1	B0/B1 Baseline Equivalence	300	0	PASS
V2	White-box search (21.4M)	60,000	0	PASS
V3	5 sabotaged + 100 mutants	105	0	PASS
V4	4-stage fact chain + 9 vectors	450	0	PASS
V5	3 OS environment profiles	7,500	0	PASS
Adv.	V1–V5 Cumulative Adv.	68,355	0	PASS
Benign	SPEC CPU2017 + Benign	50,000	0	BFDR=0
Total	Full-Spectrum Comprehensive	118,355	0	ALL PASS

B. Meta-Verification (V3)

The harness killed 100/100 mutants (Mutation Score = 1.0) and caught 5/5 sabotaged binaries, confirming discriminator sensitivity.

C. Performance & Selected SPEC CPU2017 Breakdown

Geometric mean overhead on Intel Xeon Gold 6330 was **+1.20% ($\pm 0.12\%$)**. Across a 72-hour soak test, PGM sustained an aggregate throughput of **952,991 QPS across 16 concurrent worker threads**.

TABLE IV
PGM IN-BAND LATENCY DISTRIBUTION (10^7 SAMPLES, UNIT: NS)

Metric	Min	P50	P99	Max	95% CI
C-ABI Lookup	312	353	412	620	[351, 355]
RCU Swap	390	420	438	580	[416, 424]
Fail-Closed	410	465	488	640	[462, 468]

VII. RELATED WORK & TAXONOMY

TABLE V
METHODOLOGICAL RIGOR MATRIX ACROSS DEFENSE SYSTEMS

Defense Category	Attack-Equiv	Meta-Verify	Decoupled	Oracle	V1-V5
Prompt Guardrails	Weak	None	None	No	No
MCP PEP Gateways	Partial	None	Weak	No	No
Traditional OS Sandbox	Feasible	None	Weak	No	No
Commercial Endpoint	Internal	Undisclosed	Undisclosed	No	No
Capability Kernel	Yes	Limited	Partial	No	No
DROS-PGM	Yes	100/100	Yes	PASS	PASS

Note: To the best of our survey across publicly documented literature, we did not identify an existing system reporting all five validation dimensions under a unified open package.

VIII. CONCLUSION & DECLARATIONS

This paper presented DROS-PGM, establishing a deterministic binary execution containment boundary evaluated across 118,355 executions with zero counterexamples.

ACKNOWLEDGMENT & AI COLLABORATION DISCLOSURE

In accordance with IEEE / ACM 2023+ guidelines:

- 1) **Research Originality & IP:** The system architecture, dual boundaries, formal invariants, V1–V5 methodology, and patent claims (U.S. PPA No. 64/111,973) were independently conceived and evaluated by Chun-Cheng (Jimmy) Chen.
- 2) **Role of AI Tools:** LLMs were utilized strictly as assistive productivity tools for proofreading, LaTeX typesetting, and formatting. The author retains full intellectual and legal responsibility.

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